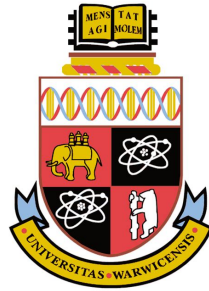


University of Warwick

Department of Economics



**UNIVERSITY
OF WARWICK**

Locked Out of the City: Housing Search Frictions and the Growth Cost of Under-Urbanisation

EC331 RESEARCH IN APPLIED ECONOMICS

Aarush Bathula

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Abstract

UK housing search frictions hold the economy below its efficient urban share, and because the urban share drives frontier productivity growth, static misallocation becomes a permanent growth loss rather than a one-time level effect. Calibrated to the UK, supply-side instruments (construction costs and matching efficiency) dominate bargaining-based corrections in both magnitude and equilibrium-selection robustness. This paper establishes these results by embedding a DMP housing search block in an endogenous-growth model and deriving a local welfare diagnostic from the free-entry envelope, which identifies the direction of inefficiency without requiring an implementable planner benchmark.

Keywords: Search and matching, housing market, urban economics, agglomeration, endogenous growth

JEL codes: E2, R23, R31, D83

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Chapter 1

Introduction

UK total factor productivity grew at 1.24 % per year on average between 1980 and 2007; between 2008 and 2016 it grew at -0.28% . Crafts and Mills (2020) describe this collapse as unprecedented in a 250-year historical series. They propose three candidate explanations: the fading ICT boom, the financial crisis and uncertainty driven by Brexit. These explanations are all macro-level shocks and do not consider spatial or structural factors. This paper proposes a complementary, structural channel: search frictions in the housing market hold the urban share below its unconstrained level, producing a sustained loss in productivity growth. The paper identifies and quantifies this channel; it does not attribute a share of the observed slowdown to it.

London recorded the highest regional output per hour in 2023, 28.5 % above the UK average (Office for National Statistics, 2025). Dense urban areas facilitate knowledge spillovers, input-sharing, and thicker labour market matching (Duranton and Puga, 2004), but relocation is not frictionless. Suitable accommodation in high-productivity regions is scarce, reflecting supply inelasticity and the time costs of matching to available units. UK rents rose 3.4 % in the year to early 2026, with English vacancy rates near 5 %, descriptive features of the housing market this paper takes as the institutional setting (Office for National Statistics, 2026). The urban share thus sits below the unconstrained level, and foregone agglomeration benefits appear as a static output loss. Hsieh and Moretti (2019, p. 2) provide the seminal analysis, showing that housing supply constraints in New York, San Francisco and San Jose reduced US GDP by 3.7% in 2009.

This paper extends that framework to a dynamic setting, where search takes time and housing is a slow-moving durable good. Urbanisation is therefore a gradual process, the level of which drives the *rate* at which the technology frontier advances. Housing frictions thereby produce not just a level loss but also a growth loss, as the economy takes longer to reach the unconstrained urban share. The importance of the growth loss is seen through the compound effect where even a 10 basis point reduction in the annual growth rate, sustained, dwarfs static losses.

The research question is therefore: when urban housing is allocated through search and slow supply adjustment, do housing-market frictions distort long-run growth as well as static spatial allocation? The paper answers this question using a dynamic search-and-matching model calibrated to the UK economy.

1.1 Main Contribution and Preview of Findings

Housing search frictions hold the urban share below its unconstrained level, and because the urban share drives frontier productivity growth, static misallocation becomes a permanent growth loss. This paper establishes that result by embedding Diamond–Mortensen–Pissarides (DMP)-style housing search within an endogenous-growth model, connecting the housing-frictions literature to agglomeration-driven growth. To the author’s knowledge, no prior paper embeds DMP-style housing search specifically in an agglomeration-driven endogenous-growth setting.

On the housing search literature, the closest antecedents are Wheaton (1990) and Head et al. (2014), whose search-theoretic treatments of the housing market characterise vacancy dynamics, transaction prices and construction without a productivity-growth channel. On the urban-productivity side, Hsieh and Moretti (2019) quantify a static output loss from housing-driven spatial misallocation using a reduced-form supply elasticity; this paper endogenises that elasticity through search and construction frictions, and identifies a dynamic growth channel absent from their framework.

The paper’s methodological move is to identify the direction of inefficiency from the free-entry envelope at the decentralised equilibrium, rather than from a planner counterfactual. Private entry drives the marginal flow-welfare effect to zero by construction, so the envelope isolates the residual growth wedge as the first-order term. This is useful in the present environment, where the global planner’s optimum implies tightness and vacancy rates well outside empirically observed ranges (Section 5.1.2) and is therefore a poor implementable benchmark. The substantive result follows: at the UK calibration, the diagnostic is positive and sign-robust across the growth-parameter space, identifying local under-urbanisation as the direction of inefficiency. The friction parameters the model identifies (matching efficiency and construction cost) map onto institutional levers in the UK: planning reform, transactional infrastructure, and transport projects that widen the effective commuting catchment of productive cities (Section 5.3.3). Results are conditional on the paper’s quasi-linear, representative-ownership setup.

1.2 Related Literature

Search-theoretic models of the housing market were built to explain outcomes internal to that market: time-to-sell, transaction prices, and the comovement of construction with vacancy. Wheaton (1990) established the stock-flow architecture the literature has worked in since. The welfare objects in this tradition are internal to the housing market – the speed at which matches form, the distribution of surplus between buyers and sellers, the extent to which prices track fundamentals.

Head et al. (2014) extend this architecture to a dynamic calibrated setting with endogenous construction and bargained prices, providing the closest methodological parallel

to Chapter 2. Their model reproduces the joint behaviour of prices and construction over the cycle; productivity is exogenous throughout, so an aggregate growth channel is absent by design. The model in Chapter 2 inherits the Wheaton stock-flow structure and the HLES dynamic calibrated envelope, and then transfers the welfare question. What the present paper asks of this architecture is not whether matches form efficiently or prices track fundamentals, but whether the urban share the architecture determines sits at a level consistent with efficient frontier growth.

Recent work has also sharpened the efficiency question within housing search itself. Gabrovski and Ortego-Marti (2019) and Gabrovski and Ortego-Marti (2026) show first that empirically plausible housing dynamics require two-sided entry, and then that in such an environment the Hosios condition alone does not restore efficiency because a within-period participation externality remains. This paper is complementary rather than redundant. It reaches the same broad conclusion that decentralised housing-market pricing is not generally efficient. However, it argues for insufficiency on a distinct margin of the growth externality.

On the spatial misallocation literature, this paper sits between the two closest modern parallels Hsieh and Moretti (2019) and Duranton and Puga (2023). The static spatial-equilibrium accounting model of Hsieh and Moretti (2019), with reduced-form local housing-supply elasticities, is well-suited to answer its question of the aggregate output loss from constrained spatial allocation without taking a stand on the microstructure of the housing market. By construction, compounding growth effects are beyond its scope. Effectively, this paper replaces the reduced-form supply elasticity with a search-based allocation friction, allowing for an analysis of the dynamic growth loss.

To construct such a growth loss, this paper builds on much of the literature surrounding agglomeration empirics. A disciplined agglomeration kernel, while still reduced-form, is necessary to identify the growth channel cleanly. Duranton and Puga (2004) produce the canonical mechanisms through which agglomeration economies arise, namely sharing, matching and learning. In Chapter 2, the agglomeration kernel relies on the microfoundations of this precedent. To similarly anchor calibrations in empirics, the paper draws on Moretti (2004) and Combes and Gobillon (2015) to establish the productive elasticity of agglomeration as positive and moderate. Rosenthal and Strange (2004) survey the agglomeration literature and establish the concavity implicit in the kernel as a stylised fact. The reduced-form implementation of agglomeration that follows is therefore the price of a tractable dynamic general equilibrium model, as opposed to an ad hoc assumption.

Extending a static misallocation model to a dynamic growth setting requires engaging the endogenous-growth literature. Endogenous growth theory establishes that aggregate growth can be sustained by externalities on non-rival inputs. Romer (1986) and Lucas (1988) are the canonical examples of this insight. Jones (1995) disciplined that tradition empirically by showing growth has not risen with research scale. This paper's growth rate

depends on a different scale variable – the urban share – and the urban share has risen with productivity in the data, so the Jones falsification does not apply. In the UK, the urban share rose alongside productivity over the post-war period through 2007 (Crafts and Mills, 2020). This is in contrast to the scale variant of much of the Jones falsification concern. The model in Chapter 2 permits growth within the admissible space. The post-2008 slowdown is on-model in that housing frictions can hold the urban share below its unconstrained level; the paper makes no claim about the share attributable to this channel.

1.3 Roadmap

The rest of the paper is structured as follows. Chapter 2 develops the model, with welfare structures developed in Chapter 3. Chapter 4 describes the calibration strategy, followed by results in Chapter 5. Chapter 6 concludes.

Chapter 2

The Model

This chapter develops a discrete-time endogenous-growth model with housing search. The economy is populated by households who can remain rural, search for urban housing, or live in the productive urban sector, and by developers who invest in housing capacity. The central friction is that access to the urban sector requires a match with a vacant unit, while the central growth channel is that the housed urban share raises frontier productivity growth through agglomeration. The chapter first describes the environment and timing, then sets out matching, values, rents and developer incentives, and finally defines equilibrium.

2.1 Environment, Agents, and Timing

Time is discrete. A unit mass of agents is distributed across three locations: rural (R), urban search (S), and housed urban (U). The economy enters each period with state $(A_{0,t}, s_t, H_t)$, where $A_{0,t}$ is frontier productivity, s_t the measure of housed urban residents, and H_t the aggregate housing stock. Physical vacancies are $V_t \equiv H_t - s_t \geq 0$.

Within each period, events occur in the following order: (i) rural households decide whether to enter urban search; (ii) searchers are matched to vacant units; (iii) matched pairs bargain over rent, then produce and consume; (iv) developers invest in new units and productivity updates. Households hold representative ownership shares in the housing stock, so rental payments are internal transfers and drop out of aggregate welfare. The framework models search for access to urban housing services and abstracts from an explicit owner-versus-renter margin.

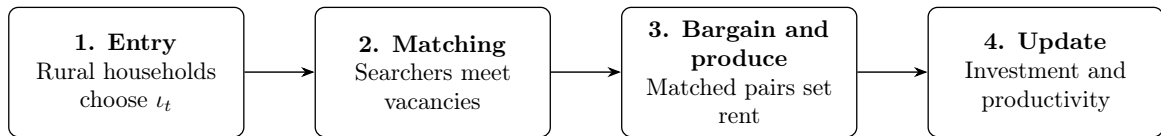


Figure 1. Within-period timing.

2.2 Housing Search, Matching, and Population Dynamics

A housed resident exits to urban search with probability $\lambda \equiv \delta_h + (1 - \delta_h)\delta$ each period. New entrants $\iota_t \in [0, 1 - s_t]$ join displaced residents to form the searcher pool $u_t = \iota_t + \lambda s_t$. Market tightness is $\theta_t \equiv V_t/u_t$. Matches form according to

$$M_t = \min\{u_t m(\theta_t), V_t\}, \quad (1)$$

where $m(\cdot)$ satisfies $m(0) = 0$, $m'(\theta) > 0$, and $m(\theta)/\theta$ strictly decreasing; these conditions ensure the vacancy fill rate $q_t \equiv M_t/V_t$ falls with tightness. The baseline specification $m(\theta) = 1 - \exp(-\mu\theta^{1-\eta})$ bounds match probabilities in $[0, 1]$ globally; at calibrated values the unconstrained regime $u_t m(\theta_t) < V_t$ holds, so the minimum does not bind. The quantitative audit verifies this slack along the reported impulse-response paths. The urban share evolves as

$$s_{t+1} = (1 - \lambda) s_t + M_t. \quad (2)$$

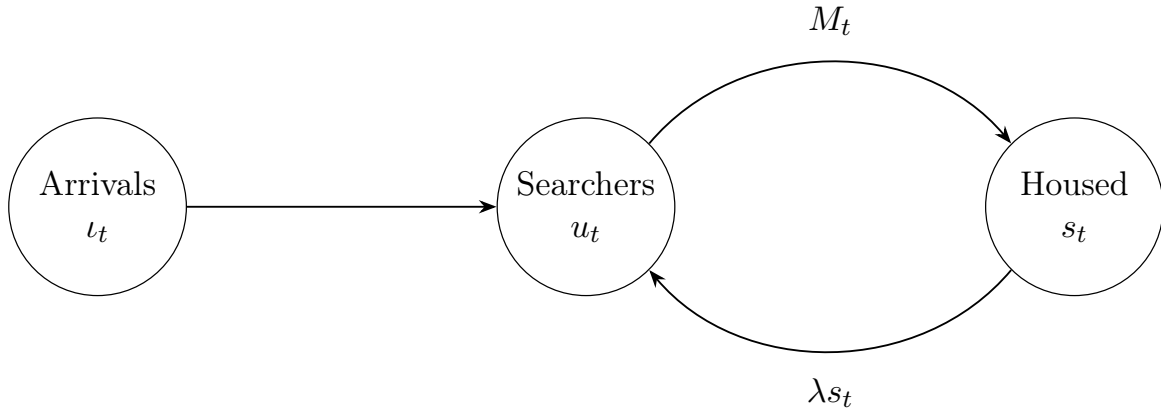


Figure 2. Population stocks and flows in the housing-search block.

2.3 Household Values, Rents, and Developer Incentives

Flow utilities are measured in efficiency units ($\tilde{x} \equiv x/A_{0,t}$) and are quasi-linear.¹ Rural agents receive $\tilde{u}_R = 1$; searchers pay a search cost $\tilde{\chi} \in (0, 1)$, so $\tilde{u}_S = 1 - \tilde{\chi}$; a housed resident at urban share s receives agglomeration surplus $z_0 s^\psi$ net of rent $\tilde{r}_t$ and congestion disutility $\tilde{\phi}_0 s$:

$$\tilde{u}_U(s, \tilde{r}) = z_0 s^\psi - \tilde{r} - \tilde{\phi}_0 s. \quad (3)$$

The searcher's value function is

$$\tilde{W}_t^S = \tilde{u}_S + \beta(1 + \nu s_t^\xi) \mathbb{E}_t[\pi_t \tilde{W}_{t+1}^U + (1 - \pi_t) \tilde{W}_{t+1}^R], \quad (4)$$

¹Quasi-linearity makes rent a transfer from tenants to representative owners, so the welfare object is allocational rather than distributional.

where $\pi_t \equiv M_t/u_t$ is the per-searcher match probability. The growth-adjusted factor $\beta^* \equiv \beta(1 + \nu s_t^\xi)$ appears because frontier growth is driven by the current urban share—equation (7) below—so a larger s_t inflates the value of every future position.

Full Bellman equations for rural and housed households and for developer asset values $\tilde{J}_t^F$ and $\tilde{J}_t^V$ are in Appendix A.

Rent is determined by Nash bargaining with tenant weight $\phi \in (0, 1)$. Because preferences are quasi-linear, the marginal effect of rent on tenant and landlord surpluses is equal and opposite, so the first-order condition reduces immediately to the surplus-sharing rule

$$\phi(\tilde{J}_t^F - \tilde{J}_t^V) = (1 - \phi)(\tilde{W}_t^U - \tilde{W}_t^S). \quad (5)$$

Developers invest $I_t \geq 0$ in new units at cost $A_{0,t} \kappa_I I_t^2/2$. Housing capacity evolves according to $H_{t+1} = (1 - \delta_h)H_t + I_t$, so new construction offsets depreciation only gradually. The interior investment condition equates the marginal cost of supply to the discounted value of a future vacant unit:

$$\kappa_I I_t = \beta(1 + \nu s_t^\xi) \mathbb{E}_t[\tilde{J}_{t+1}^V]. \quad (6)$$

2.4 Growth Externality and Resource Constraint

Frontier productivity follows

$$A_{0,t+1} = A_{0,t}(1 + \nu s_t^\xi), \quad \nu \geq 0, \quad \xi \in (0, 1], \quad (7)$$

so the balanced-growth-path (BGP) growth rate $g^* = \nu(s^*)^\xi$ is determined endogenously by the equilibrium urban share. The convergence condition $\beta(1 + \nu) < 1$ ensures detrended value functions are well-defined. Equation (7) is a reduced-form agglomeration-driven growth channel: ξ is the elasticity of the growth rate with respect to the urban share, ν its scale. Rural agents each produce one efficiency unit; a housed resident at share s produces $z_0 s^\psi$ through agglomeration, giving aggregate output $\tilde{Y}(s) = (1 - s) + z_0 s^{1+\psi}$. The resource constraint is

$$\tilde{Y}(s_t) = \tilde{C}_t + \frac{\kappa_I}{2} I_t^2 + \tilde{\gamma}_V V_t, \quad (8)$$

where $\tilde{\gamma}_V > 0$ is the per-period carrying cost of a vacant unit. Walras' Law under representative ownership ensures equation (8) holds in equilibrium.

These pieces together produce the chapter's central mechanism. Reduced matching frictions (higher μ) or lower construction costs (lower κ_I) raise the expected value of a vacant unit, inducing more investment through equation (6). Greater capacity expands vacancies V_{t+1} , tightens the market, and raises match formation M_{t+1} . The law of motion in equation (2) then translates additional matches into a larger housed urban share s_{t+1} , which feeds directly into the growth equation (7). Static search frictions therefore have

dynamic growth consequences: on the balanced growth path, they permanently lower the frontier growth rate.

2.5 Equilibrium Definition

Definition 1 (Competitive equilibrium). Given $(A_{0,0}, s_0, H_0)$ with $H_0 \geq s_0$, a *competitive equilibrium* is a sequence of quantities $\{\iota_t, u_t, V_t, \theta_t, M_t, s_{t+1}, \tilde{r}_t, I_t, H_{t+1}, \tilde{C}_t\}_{t \geq 0}$ and detrended values $\{\tilde{W}_t^k, \tilde{J}_t^\ell\}$ satisfying, for all t : (i) the household Bellman equations and entry rule (Appendix A); (ii) matching and population dynamics in equations (1)–(2); (iii) the Nash rent condition in equation (5) and developer investment condition in equation (6); (iv) the growth process in equation (7) and the resource constraint in equation (8).

A *BGP* is a competitive equilibrium in which $(s_t, H_t, V_t, \theta_t, I_t)$ are constant and $A_{0,t}$ grows at rate $g^* = \nu(s^*)^\xi$. Stationarity of equation (2) requires $M^* = \lambda s^*$; stationarity of the housing stock requires $I^* = \delta_h H^*$. The full BGP system in detrended unknowns is given in Appendix A.

Entry is governed by a comparison of continuation values: a household moves into search when $\tilde{W}_t^S \geq \tilde{W}_t^R$. Interior entry requires the knife-edge condition $\tilde{W}^{S,*} = \tilde{W}^{R,*}$; existence and uniqueness are shown in Appendix A.

Chapter 3

Welfare and the Local Diagnostic

The competitive equilibrium of Chapter 2 is generically inefficient. This chapter identifies why, defines the objects used to measure the inefficiency, and states the testable implication that Chapter 5 quantifies. Full derivations are in Appendix A.

Under representative ownership, rents are internal transfers and per-period detrended welfare is

$$\tilde{\mathcal{W}}^* \equiv \tilde{C}^* - \tilde{\chi} u^* - \tilde{\phi}_0 (s^*)^2, \quad (9)$$

so lifetime welfare on the balanced growth path is $\mathcal{V}^* = \tilde{\mathcal{W}}^*/(1 - \beta^*)$, where $\beta^* \equiv \beta(1 + \nu(s^*)^\xi)$ is the growth-adjusted discount factor.

3.1 Two Externalities

Private entry decisions ignore two effects. The first is *search congestion*: a marginal entrant raises the searcher pool u_t , lowers market tightness θ_t , and reduces every other searcher's match probability. The second is a *growth externality*: a marginal housed resident raises s_{t+1} , and hence the growth factor νs_{t+1}^ξ in equation (7), for every agent in every future period.

The two externalities act on different margins. Search congestion operates within a period, on the match-formation margin; the growth externality operates across periods, on the stock of housed residents that drives $A_{0,t}$. The two are therefore not substitutes: correcting one does not correct the other.

3.2 Hosios as a Partial Correction

Let $\varepsilon(\theta) \equiv \theta m'(\theta)/m(\theta)$ denote the matching-function elasticity. The search congestion externality is internalised when the tenant bargaining weight satisfies

$$\phi = 1 - \varepsilon(\theta^*). \quad (10)$$

This is the standard Hosios efficiency condition (Hosios, 1990). The derivation, which compares the decentralised free-entry condition against the planner's marginal-entry condition at fixed vacancies, is standard for DMP environments and is carried out in Appendix A.

When $\nu > 0$, the Hosios condition alone does not restore efficiency. The growth externality remains, because equation (10) corrects only the within-period congestion

margin; it does not internalise the effect of s_{t+1} on future productivity growth. This logic is consistent with the housing-search efficiency note of Gabrovski and Ortego-Martí (2026), but here the inefficiency is sharpened by the endogenous-growth channel. Quantifying the residual growth wedge is the subject of the next section.

3.3 The Local Welfare Diagnostic

Define the *growth wedge*

$$\Omega^g(s) \equiv \beta \nu \xi s^{\xi-1} \mathcal{V}^*(s), \quad (11)$$

which captures the marginal social value of raising the urban share through the growth channel. The *local welfare wedge* at the decentralised allocation is then

$$D(s_{CE}^*) \equiv \frac{d\tilde{\mathcal{W}}^*(s_{CE}^*)}{ds} + \Omega^g(s_{CE}^*). \quad (12)$$

A positive $D(s_{CE}^*)$ means a marginal increase in the urban share raises lifetime welfare: the growth dividend exceeds the marginal flow-welfare cost. The paper's headline welfare claim is that $D(s_{CE}^*) > 0$ at the calibration of Chapter 4, and is robust to the growth parameters (ν, ξ) .

Remark 1 (Envelope). The envelope neutralises the marginal welfare effect of search frictions, not their role in setting the level; the DMP block remains structurally necessary because the equilibrium urban share is endogenously pinned down by it. At the decentralised allocation, private free entry drives $d\tilde{\mathcal{W}}^*/ds$ toward zero by construction. The near-zero value of the flow slope at s_{CE}^* is therefore an envelope consequence of free entry, not evidence that the search externality is quantitatively small. The search distortion instead appears through the *level* of the decentralised allocation in a full-system Hosios counterfactual, which is a separate object from the marginal diagnostic in equation (12). $D(s_{CE}^*)$ should be read as isolating the marginal growth wedge at the observed competitive equilibrium, not as a decomposition of total inefficiency.

3.4 Conditional Interpretation

The diagnostic should therefore be read within the transferable-utility scope stated in Chapter 2: it is a marginal test of whether the calibrated competitive equilibrium is locally under-urbanised, not a full welfare decomposition or implementable optimal policy.

Chapter 4

Calibration to the UK economy

The calibration proceeds in two stages. External parameters (Table 1) are set from the literature; three balanced-growth-path moments (Table 2) then identify the remaining three parameters in closed form.

Table 1. Calibrated parameters

Parameter	Symbol	Value	Source / Identification
<i>Externally set</i>			
Discount factor	β	0.96	Cooley and Prescott (1995)
Housing depreciation	δ_h	0.02	Fraumeni (1997); Miles and Monro (2019)
Match separation	δ	0.05	English Housing Survey (2022) (all-tenure weighted)
Matching elasticity	η	0.50	Petrongolo and Pissarides (2001)
Tenant bargaining weight	ϕ	0.50	Symmetric Nash baseline
Agglomeration curvature	ψ	0.05	Combes and Gobillon (2015)
Growth scale	ν	0.014	Bank of England (2018), 1980–2007 avg.
Growth elasticity	ξ	0.70	Reduced-form concavity
Vacancy carrying cost	$\tilde{\gamma}_V$	0.010	Not targeted auxiliary parameter
Search cost	$\tilde{\chi}$	0.020	Not targeted auxiliary parameter
Congestion disutility	$\tilde{\phi}_0$	0.100	Not targeted auxiliary parameter
<i>Internally calibrated</i>			
Matching efficiency	μ	0.654	Tightness target $\theta^* = 0.35$ via $m(\theta^*)$
Agglom. productivity	z_0	1.153	Free entry $\tilde{W}^{S,*} = \tilde{W}^{R,*}$ at $s^* = 0.85$
Construction cost	$\bar{\kappa}_I$	51.45	Investment FOC at implied $\tilde{J}^{V,*}$

Note. Internally calibrated values are recovered exactly from the three moment conditions; the appendix reports the corresponding closed-form expressions and implied BGP quantities.

Each target in Table 2 identifies one internally calibrated parameter through a single equilibrium condition: the urban share $s^* = 0.85$ (World Bank, 2026) pins down z_0 through free-entry indifference; the vacancy-to-stock ratio $V^*/H^* = 0.07$ (English Housing Survey, 2025) pins down $\bar{\kappa}_I$ through the investment first-order condition; market tightness $\theta^* = 0.35$ pins down μ through inversion of equation (1). Because tightness is not directly observed in UK housing data, I treat θ^* as a reduced-form BGP target consistent with the vacancy share and the model’s matching identities. The World Bank classification is conservative for the model’s interpretation of “productive urban”: narrower measures (e.g. top- N metropolitan share) would push s^* lower, away from the asymptote, and mechanically increase the growth wedge Ω^g . Recovered values are reported in Table 1.

Table 2. Calibration targets

Moment	Target	Identifies	Source
Market tightness	$\theta^* = 0.35$	μ	Vacancy share and matching identities at BGP
Urban population share	$s^* = 0.85$	z_0	World Bank (2026)
Vacancy-to-stock ratio	$V^*/H^* = 0.07$	$\bar{\kappa}_I$	English Housing Survey (2025) + short-term voids

Note. The target combines the EHS 2024–25 headline estimate of approximately 5% vacant dwellings with a 2 percentage point allowance for frictional vacancies between tenancies or undergoing minor works. The model is tenure-agnostic and counts all unoccupied units regardless of administrative classification.

The growth elasticity $\xi = 0.70$ encodes diminishing returns of frontier growth to urbanisation; the linear limit $\xi = 1$ is inconsistent with the empirical pattern of declining agglomeration returns at high densities (Rosenthal and Strange, 2004; Combes and Gobillon, 2015), and across $\xi \in [0.5, 1]$ the implied growth rate varies only from 1.18% to 1.28%, leaving the welfare decomposition qualitatively unchanged. The growth scale $\nu = 0.014$ is set so the calibrated CE reproduces the 1980–2007 UK TFP trend, loading that trend entirely onto the agglomeration–urbanisation channel and abstracting from other TFP drivers. The calibrated ν should therefore be read as an upper-bound calibration of the channel rather than a point estimate; Table 5 (Panel B) confirms $D(s_{CE}^*) > 0$ across $\nu \in [0.010, 0.025]$, so the directional conclusion survives substantial reductions in assumed magnitude. The auxiliary parameters $\tilde{\gamma}_V$, $\tilde{\chi}$, and $\tilde{\phi}_0$ are externally set rather than targeted, with baseline values chosen to have plausible flow-utility interpretations as fractions of annual frontier productivity.

Appendix B collects the closed-form recovery of $(\mu, z_0, \bar{\kappa}_I)$, the implied BGP quantities, and the sensitivity checks for ξ , $\tilde{\gamma}_V$, $\tilde{\chi}$, and $\tilde{\phi}_0$.

The calibrated bargaining weight $\phi = 0.50$ leaves the search-congestion externality partially uncorrected, with a Hosios gap of $\phi - [1 - \varepsilon(\theta^*)] \approx -0.091$.

Chapter 5

Quantitative Results

This chapter reports the calibrated Balanced Growth Path, local welfare diagnostic $D(s_{CE}^*)$ defined in Section 3.3, and dynamic verification of the comparative-static signs of the friction parameters through impulse responses. The chapter closes with policy implications from this paper.

5.1 Calibrated Equilibrium and Welfare Diagnostic

5.1.1 Features of the calibrated UK equilibrium

Table 3 reports the outputs from the calibrated BGP. The urban share s^* is already high, implying that the growth channel operates near its asymptote. For an already urbanised economy like the UK, this means that marginal movements in s^* produce modest changes in the growth rate. The matching market is slack at $\theta^* = 0.35$, translating to plenty of vacancies relative to searchers, muting the congestion externality discussed in Section 3.1 at the margin. With $\phi = 0.50 < 1 - \varepsilon(\theta^*) \approx 0.59$, tenants capture less surplus than their marginal contribution to match formation, so private entry is too low relative to the congestion-correcting benchmark. The UK calibration therefore lands on the under-entry side.

The residual search-congestion distortion and the growth wedge point in the same direction: both favour higher urbanisation. The local diagnostic should not, however, be read as a clean decomposition of the search margin into congestion, thickness, and composition effects. Because free entry pins down the marginal search condition, the envelope property in Section 3.3 absorbs the local flow effect of search frictions; the remaining first-order term is the growth wedge evaluated at the decentralised allocation. This composition is structural. The free-entry condition admits two roots at baseline; the calibrated results select the high-urbanisation branch throughout.¹

¹Appendix A discusses equilibrium multiplicity and the selection criterion.

Table 3. Calibrated BGP ($\xi = 0.70$)

Quantity	Value
Urban share s^*	0.8500
Growth rate $g^* = \nu(s^*)^\xi$ (% p.a.)	1.2400
Housing stock H^*	0.9140
Tightness θ^*	0.3500
Vacancy elasticity $\varepsilon(\theta^*)$	0.4095
Hosios gap $\phi - [1 - \varepsilon(\theta^*)]$	-0.0905

5.1.2 Local welfare diagnostic

Evaluating the local diagnostic of equation (12) at the calibration yields:

Proposition 1 (Local inefficiency at the calibrated BGP). *At the calibrated BGP with $\nu > 0$ and $\xi \in (0, 1]$, the numerical welfare wedge satisfies $D(s_{CE}^*) > 0$. The decentralised equilibrium is therefore locally inefficient in the direction of under-urbanisation.*

Table 4. Local welfare diagnostic at s_{CE}^* ($\xi = 0.70$)

Object	Value
Flow welfare slope $d\tilde{\mathcal{W}}^*/ds$	1.4×10^{-3}
Growth wedge $\Omega^g = \beta\nu\xi s^{\xi-1}\mathcal{V}^*$	0.3617
Local welfare wedge $D(s_{CE}^*)$	0.3631

Table 4 reports $D(s_{CE}^*) = 0.363$, the chapter's headline welfare finding. The envelope caveat of Section 3.3 governs the interpretation of the flow slope.

For reference, the global planner benchmark yields $s_{SP}^* = 0.9707$ and a welfare gap $\Delta = 3.6379\%$, but this should be read as an upper bound on the welfare cost rather than an implementable policy target. The efficient urbanisation benchmark requires enough slack capacity to keep search matches available while moving nearly the whole population into the urban sector, which is why it implies tightness $\theta_{SP}^* \approx 3.31$ and a vacancy rate of approximately 25%, well outside empirically observed ranges. This motivates the focus on the local diagnostic, which identifies the direction and source of inefficiency at the decentralised allocation.

5.1.3 Evidence for the envelope interpretation

Table 5. Sensitivity of the local welfare diagnostic to growth parameters

	$D(s_{CE}^*)$	Ω^g	$d\tilde{W}^*/ds$	s_{SP}^*	Δ (%)
<i>Panel A: Varying growth concavity ξ</i>					$(\nu = 0.014)$
$\xi = 0.50$	0.2717	0.2706	0.0011	0.9654	2.50
$\xi = 0.60$	0.3185	0.3173	0.0013	0.9684	3.07
$\xi = 0.70$	0.3631	0.3617	0.0014	0.9707	3.64
$\xi = 0.80$	0.4055	0.4040	0.0016	0.9726	4.20
$\xi = 0.90$	0.4459	0.4442	0.0017	0.9742	4.77
$\xi = 1.00$	0.4843	0.4825	0.0019	0.9756	5.33
<i>Panel B: Varying growth scale ν</i>					$(\xi = 0.70)$
$\nu = 0.010$	0.2356	0.2329	0.0027	0.9644	2.17
$\nu = 0.014$	0.3670	0.3656	0.0014	0.9709	3.68
$\nu = 0.018$	0.5350	0.5351	-0.0001	0.9755	5.79
$\nu = 0.022$	0.7573	0.7591	-0.0018	0.9792	8.83
$\nu = 0.025$	0.9778	0.9810	-0.0032	0.9816	12.12

Note. Calibration targets $(s^*, V^*/H^*, \theta^*)$ held fixed across panels; recovered parameters $(\mu, z_0, \bar{\kappa}_I)$ adjust at each value.

Table 5 confirms that $D(s_{CE}^*) > 0$ and is driven by Ω^g rather than the flow slope across the growth-parameter sweep. Sign-invariance is the structural claim; the quantitative magnitude scales with growth parameters. Additional welfare-diagnostic robustness checks are reported in Appendix C.

5.2 Dynamic Comparative Statics

Impulse responses trace how each friction parameter propagates through the mechanisms of Chapter 2. The shocks are normalised as one-per-cent log innovations to the matching-efficiency, construction-cost, and vacancy-carrying-cost states, each following an AR(1) process with persistence $\rho = 0.8$. The IRFs verify the comparative-static signs of Propositions 3–5 (Appendix A), and reveal the horizon over which the growth externality operates.

5.2.1 Matching-efficiency shock (ε^μ)

A shock to matching efficiency is the cleanest isolation of the paper’s mechanism. A friction that operates entirely on the match margin, it produces a growth response that persists for decades, because the housed population accumulates slowly (Appendix A).

The fill rate q_t rises $\approx 1\%$ on impact, boosting per-searcher matches. Entry overshoots on impact, then crashes from $+0.65\%$ at $t = 1$ to -0.9% at $t = 2$, as the free-entry condition re-establishes $\tilde{W}_t^S = \tilde{W}_t^R$. Market tightness falls as the searcher pool absorbs the shock, translating to a 40-period persistent response in s_t and g_t . Welfare flow $\tilde{W}_t$ dips briefly at $t = 1$ ($\approx -3 \times 10^{-3}\%$) before rising: searchers incur the flow cost $\tilde{\chi}$ on

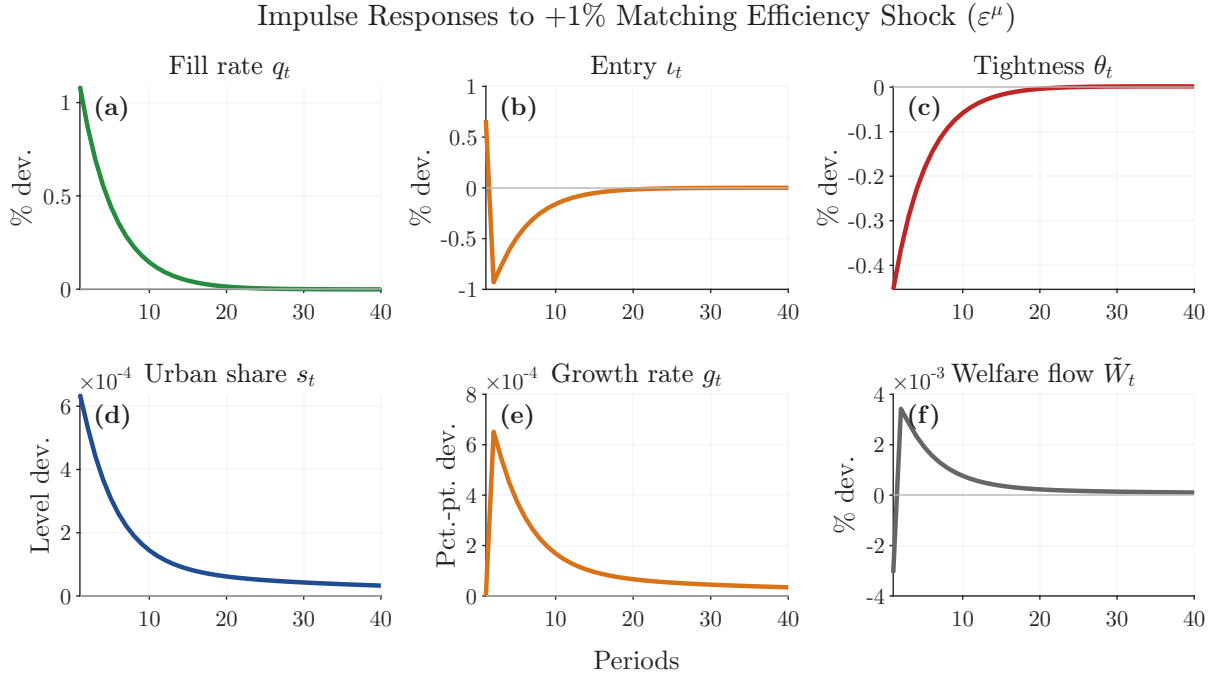


Figure 3. Impulse responses to a matching-efficiency shock.

impact, but the resulting higher urban share generates productivity growth that dominates cumulatively.

5.2.2 Construction-cost shock (ε^{κ_I})

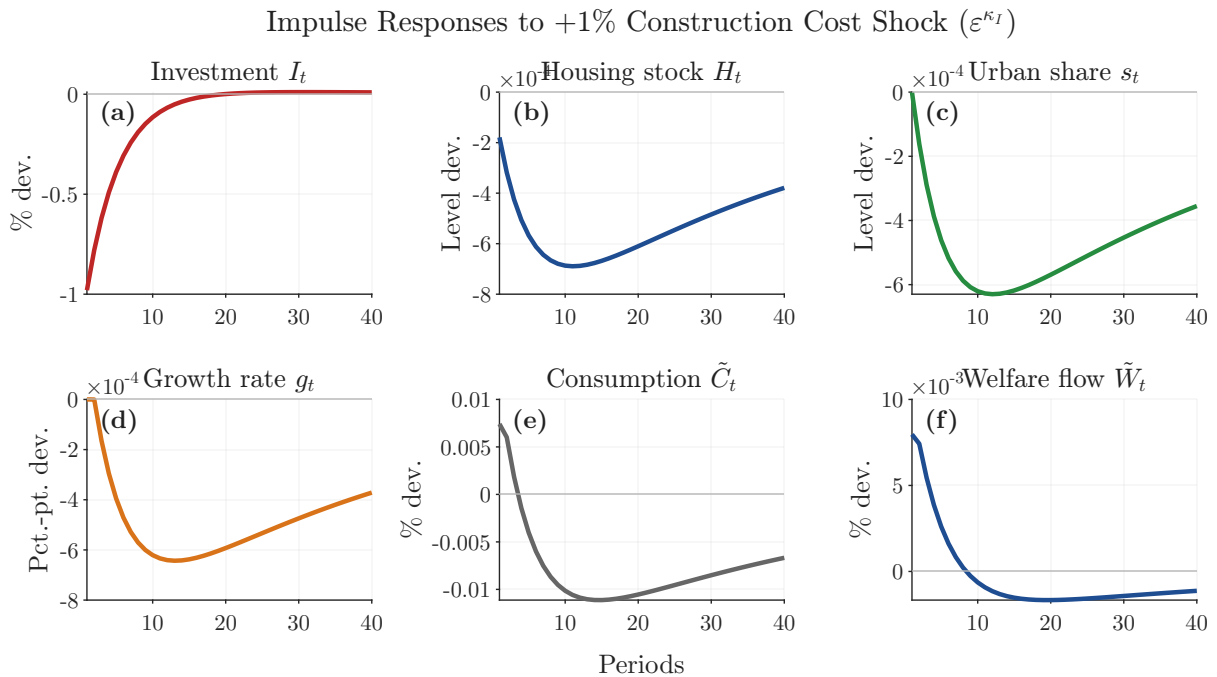


Figure 4. Impulse responses to a construction-cost shock.

Investment falls via the FOC in equation (6); H_t accumulates slowly because investment cannot offset depreciation initially, with a trough at $t = 11$. In response, rents rise along

with vacancy values. One can observe the intuitive fall in s_t and g_t , with s_t reaching its trough at $t = 12$ and g_t at $t = 13$, and with neither measure recovering by the end of the horizon $t = 40$. Consumption $\tilde{C}_t$ rises on impact by approximately +0.007% — a consequence of the resource constraint in equation (8): less output is diverted to the quadratic investment cost $\kappa_I I_t^2/2$, mechanically freeing consumption. This is not a welfare gain channel; it is resource reshuffling within a constraint, and $\tilde{W}_t$ captures the difference correctly through the congestion and agglomeration terms.

Welfare flips sign around $t = 9$ as the accumulated growth loss dominates the impact consumption gain.

5.2.3 Vacancy-cost shock ($\varepsilon^{\tilde{V}}$)

Impulse Responses to +1% Vacancy Carrying Cost Shock (ε^{γ_V})

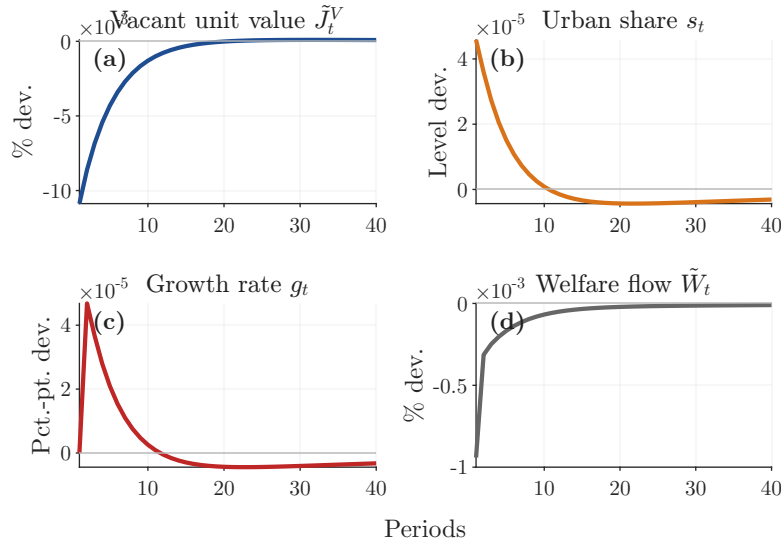


Figure 5. Impulse responses to a vacancy-carrying-cost shock.

The vacancy-cost shock propagates through the developer side: higher carrying costs sharply lower the vacant-unit asset value, absorbed through rents. The searcher pool increases, boosting s_t and g_t , but the magnitude is an order smaller than the construction and matching-efficiency shocks. As the unit values and rents recover, urbanisation falls, with a sign-flip after $t \approx 10$ that does not recover by period 40.

5.2.4 Robustness

The dynamic mechanisms of how $\mu, \kappa_I, \tilde{\gamma}_V$ impact urbanisation and growth in the model are structurally invariant to the growth calibration. The IRF fans provide evidence of this, with the baselines sitting inside the envelopes.

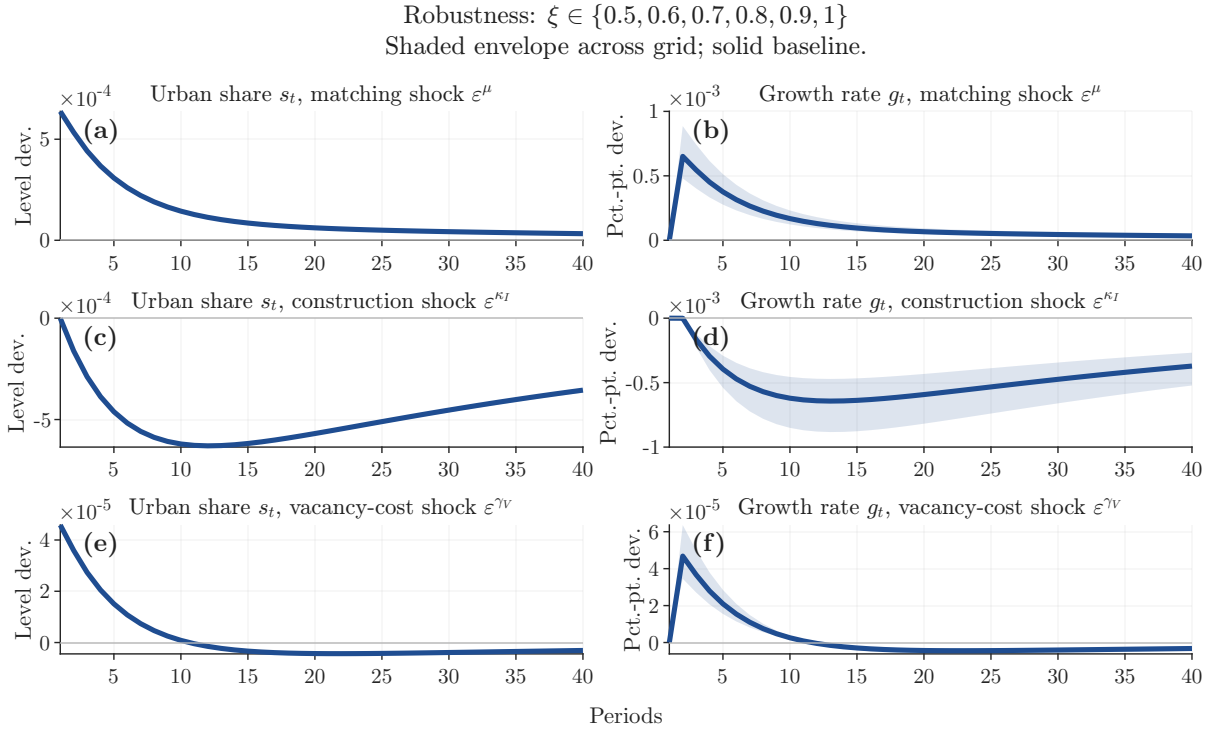


Figure 6. IRF robustness to variation in the urban-share elasticity ξ .

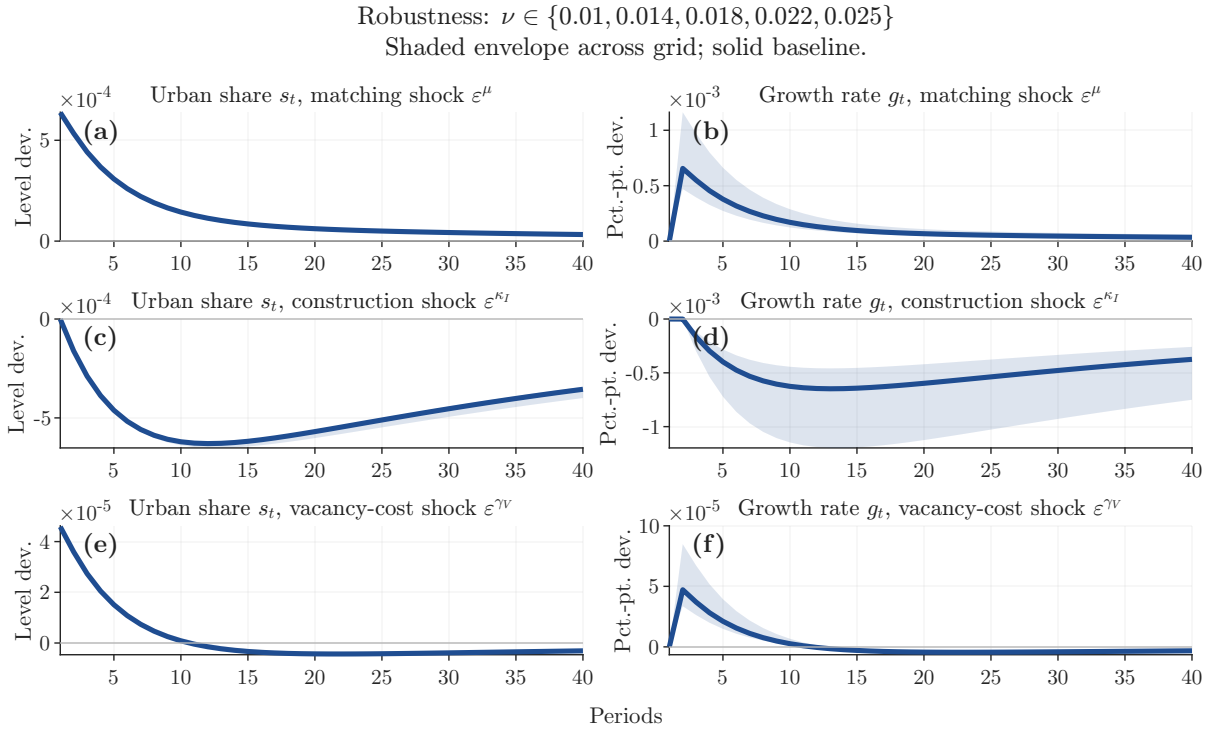


Figure 7. IRF robustness to variation in the growth scale ν .

5.3 Policy Implications

This section synthesises the welfare analysis of Section 5.1 and the dynamic comparative statics of Section 5.2 to defend a model-internal directional claim: supply-side instruments (μ, κ_I) dominate bargaining-based corrections (ϕ -shifts). The dominance rests on exactly two structural reasons.

5.3.1 Magnitude

A one-per-cent shock to μ or κ_I moves the urban share by an order of magnitude more than the same shock to $\tilde{\gamma}_V$ (Section 5.2). ϕ -adjustments cannot reach the matching or construction margins: they operate through the surplus-sharing rule and enter the quantity block only through $\tilde{J}^V$ – the channel the vacancy-cost shock isolates, and the weakest in the IRF ranking. Bargaining-based corrections therefore travel through the smallest-magnitude transmission route in the model, while supply-side corrections act on the largest two.

5.3.2 Equilibrium-selection robustness

The interior BGP region in ϕ -space contracts under tighter supply calibrations. At baseline, $\phi = 0.50$ sits comfortably inside the admissible region; with κ_I doubled, the upper boundary of the high-urbanisation branch tightens to $\phi \in [0.531, 0.532]$, beyond which the high- s root disappears entirely and the economy collapses onto the low-urbanisation trap identified in Remark 2 in Appendix A. Bargaining-based reform therefore exposes policymakers to a multiplicity-driven coordination failure: under unfavourable supply conditions, a small misjudgement of ϕ is enough to tip the economy off its high-urbanisation branch. Supply-side reform faces no symmetric risk—lowering κ_I or raising μ widens the admissible ϕ -region rather than narrowing it.

The mapping from κ_I and μ to UK policy levers is institutional rather than algebraic.

5.3.3 Operationalising variables

The construction-cost parameter is the model’s reduced-form wedge between physical building costs and the cost of adding urban housing capacity. In the UK, this wedge is dominated by the planning system: discretionary refusal risk, the green belt, Section 106 obligations, and the time-to-decision under the Town and Country Planning framework. Hilber and Vermeulen (2016) identify English regulatory restrictiveness as the binding constraint on local housing supply; Glaeser and Gyourko (2018) reach an analogous conclusion for productive US metros. Reforms that lower κ_I act on this wedge: green-belt liberalisation, by-right zoning in designated growth zones, reductions in Section 106 burdens, and statutory deadlines on local-authority decisions.

The matching-efficiency parameter governs how quickly a vacant urban unit becomes occupied. Three institutional channels move μ : search infrastructure (open property registers, rental platforms, standardised tenancy data); transactional friction (conveyancing

time, security-of-tenure rules, and stamp-duty thresholds); and transport, where any project widening a productive city’s commuting catchment enlarges the set of reachable vacant units without local relocation. Heblich et al. (2018) document this for nineteenth-century London commuter rail. HS2 Phase 1 is the closest contemporary UK analogue; the structural claim is that infrastructure widening the effective commuting match surface is a μ -instrument.

5.3.4 Caveat

The ranking is conditional on the paper’s quasi-linear, representative-ownership setup. Under non-transferable utility or heterogeneous ownership, ϕ would re-enter as a redistributive lever and the ordering could differ. The local diagnostic identifies directional dominance under this scope, not the magnitude of a full ϕ -counterfactual. A sufficiently large adverse shock could push the economy toward the low-urbanisation root identified in Appendix A; quantifying the basin of attraction is left for future work. Expansion to multi-region productivity would also introduce worthwhile spatial redistribution.

5.3.5 Compounding

Figure 8 illustrates why even modest movements in the friction parameters matter dynamically. A 10 basis-point persistent wedge in the BGP growth rate compounds to a 4.03% productivity gap by year 40. This is not a solved transition path or a permanent-reform simulation; it is a growth-accounting illustration of why the static-vs-dynamic distinction is the one that matters for policy ranking in this model.

5.3.6 Falsifiability

The mechanism is empirically distinguishable from static misallocation: conditional on urban share, regions with lower housing-search frictions (e.g., planning restrictiveness Hilber and Vermeulen (2016)) should show higher TFP growth; a null would refute it.

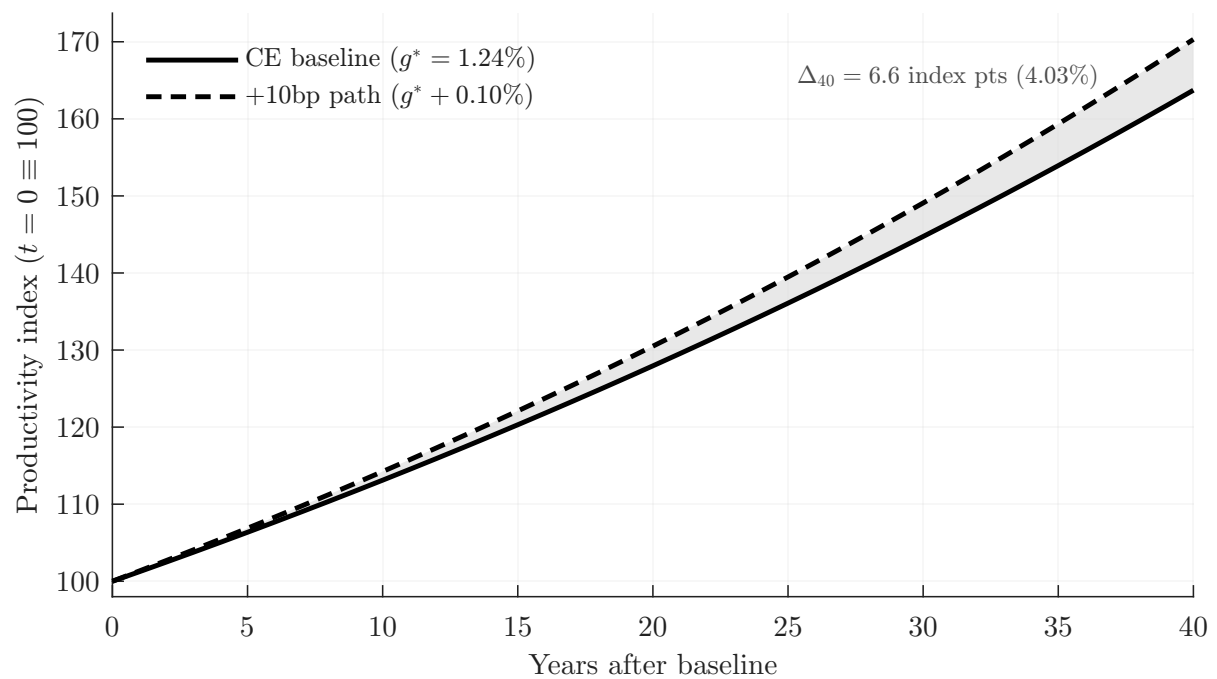


Figure 8. Growth-accounting illustration of a small persistent growth wedge. The baseline path compounds at the calibrated decentralised BGP growth rate $g_{CE}^* = 1.24\%$; the alternative path adds 10 basis points, showing how a small persistent growth wedge cumulates into a visible productivity gap over policy-relevant horizons. Both are indexed to 100 at $t = 0$. This figure is not a solved transition path.

Chapter 6

Conclusion

This dissertation has asked whether housing-market frictions distort long-run growth as well as static spatial allocation when access to high-productivity urban locations is governed by search and slow housing adjustment. The answer is yes. By holding the urban share below its unconstrained level, housing search frictions turn a one-time level loss into a permanent growth loss.

The paper establishes this by embedding a Diamond–Mortensen–Pissarides housing-search block in an endogenous-growth environment: Hosios can correct within-period search congestion, but not the growth externality from urbanisation. At the UK calibration, the local welfare diagnostic $D(s_{CE}^*)$ is positive and sign-robust, identifying under-urbanisation without requiring an implementable planner benchmark. Quantitatively, matching efficiency μ and construction cost κ_I dominate the vacancy-carrying cost $\tilde{\gamma}_V$, while bargaining-based reform faces an equilibrium-selection risk that supply-side reform avoids.

These conclusions are conditional on a quasi-linear, representative-ownership setup with a tenure-agnostic housing block. With income effects or heterogeneous ownership, rents would no longer be welfare-neutral transfers and ϕ would re-enter as a redistributive lever. The supply-side ranking is therefore about allocational efficiency under transferable utility, not distributional welfare.

The natural extension is a heterogeneous-agent model separating owner-occupiers, social tenants, and private renters, with matching differentiated by tenure. The welfare object would become a distribution of consumption-equivalent gains rather than a representative-agent diagnostic.

Once housing is treated as a frictional allocation mechanism rather than a reduced-form supply elasticity, the welfare economics of spatial misallocation is a growth problem as well as a level problem.

Appendix A

Model Details, Derivations, and Supplementary Theory

A.1 Value Functions and Omitted Equilibrium Conditions

The reduced-form statement in Chapter 2 suppresses the full dynamic system. In detrended efficiency units, the household Bellman equations are

$$\tilde{W}_t^R = 1 + \beta(1 + \nu s_t^\xi) \mathbb{E}_t[\max\{\tilde{W}_{t+1}^R, \tilde{W}_{t+1}^S\}], \quad (13)$$

$$\tilde{W}_t^S = 1 - \tilde{\chi} + \beta(1 + \nu s_t^\xi) \mathbb{E}_t[\pi_t \tilde{W}_{t+1}^U + (1 - \pi_t) \max\{\tilde{W}_{t+1}^S, \tilde{W}_{t+1}^R\}], \quad (14)$$

$$\tilde{W}_t^U = z_0 s_t^\psi - \tilde{r}_t - \tilde{\phi}_0 s_t + \beta(1 + \nu s_t^\xi) \mathbb{E}_t[(1 - \lambda) \tilde{W}_{t+1}^U + \lambda \tilde{W}_{t+1}^S]. \quad (15)$$

The entry rule follows from the comparison of search and rural continuation values:

$$\iota_t = \begin{cases} 0, & \tilde{W}_t^S < \tilde{W}_t^R, \\ \text{any value in } (0, 1 - s_t), & \tilde{W}_t^S = \tilde{W}_t^R, \\ 1 - s_t, & \tilde{W}_t^S > \tilde{W}_t^R. \end{cases}$$

Developer asset values are

$$\tilde{J}_t^F = \tilde{r}_t + \beta(1 + \nu s_t^\xi)(1 - \delta_h) \mathbb{E}_t[(1 - \delta) \tilde{J}_{t+1}^F + \delta \tilde{J}_{t+1}^V], \quad (16)$$

$$\tilde{J}_t^V = -\tilde{\gamma}_V + \beta(1 + \nu s_t^\xi)(1 - \delta_h) \mathbb{E}_t[q_t \tilde{J}_{t+1}^F + (1 - q_t) \tilde{J}_{t+1}^V]. \quad (17)$$

Writing $\tilde{J}_t^F = \tilde{r}_t + \hat{J}_t^F$ and $\tilde{W}_t^U = \hat{W}_t^U - \tilde{r}_t$, the Nash condition in Chapter 2,

$$\phi(\tilde{J}_t^F - \tilde{J}_t^V) = (1 - \phi)(\tilde{W}_t^U - \tilde{W}_t^S),$$

implies the rent formula

$$\tilde{r}_t = (1 - \phi)(\hat{W}_t^U - \tilde{W}_t^S) - \phi(\hat{J}_t^F - \tilde{J}_t^V). \quad (18)$$

A.2 Balanced Growth Path System

The stationary investment condition becomes

$$\kappa_I I^* = \beta^* \tilde{J}^{V,*}. \quad (19)$$

With interior entry, the deterministic BGP system is:

The admissible interior branch also requires $M^* < V^*$, equivalently $q^* = M^*/V^* < 1$.

Interior entry exists uniquely. Fix candidate BGP stocks (s^*, H^*) satisfying the stock-flow identities and the investment condition (19). Then $\tilde{W}^{R,*} = 1/(1 - \beta^*)$ is independent of ι^* , while higher entry raises $u^* = \iota^* + \lambda s^*$, lowers $\theta^* = V^*/u^*$, and reduces $\pi^* = m(\theta^*)$. It follows from (14) that $\tilde{W}^{S,*}$ is continuous and strictly decreasing in ι^* .

Table 6. BGP system in the interior equilibrium

Unknown	BGP condition	Source
s^*	$M^* = \lambda s^*$	Equation (2) in Chapter 2
H^*	$I^* = \delta_h H^*$	Stationary housing law of motion
V^*	$V^* = H^* - s^*$	Vacancy identity
u^*	$u^* = \iota^* + \lambda s^*$	Searcher identity
θ^*	$\theta^* = V^*/u^*$	Tightness definition
M^*	$M^* = u^* m(\theta^*)$	Unconstrained matching branch
π^*	$\pi^* = m(\theta^*)$	From $M^* = u^* m(\theta^*)$
q^*	$q^* = M^*/V^*$	Fill-rate definition
$\tilde{W}^{R,*}$	$\tilde{W}^{R,*} = 1 + \beta^* \tilde{W}^{R,*}$	(13)
$\tilde{W}^{S,*}$	$\tilde{W}^{S,*} = 1 - \tilde{\chi} + \beta^* [\pi^* \tilde{W}^{U,*} + (1 - \pi^*) \tilde{W}^{R,*}]$	(14)
$\tilde{W}^{U,*}$	$\tilde{W}^{U,*} = z_0(s^*)^\psi - \tilde{r}^* - \tilde{\phi}_0 s^* + \beta^* [(1 - \lambda) \tilde{W}^{U,*} + \lambda \tilde{W}^{S,*}]$	(15)
$\tilde{J}^{F,*}$	$\tilde{J}^{F,*} = \tilde{r}^* + \beta^* (1 - \delta_h) [(1 - \delta) \tilde{J}^{F,*} + \delta \tilde{J}^{V,*}]$	(16)
$\tilde{J}^{V,*}$	$\tilde{J}^{V,*} = -\tilde{\gamma}_V + \beta^* (1 - \delta_h) [q^* \tilde{J}^{F,*} + (1 - q^*) \tilde{J}^{V,*}]$	(17)
$\tilde{r}^*$	$\phi(\tilde{J}^{F,*} - \tilde{J}^{V,*}) = (1 - \phi)(\tilde{W}^{U,*} - \tilde{W}^{S,*})$	Equation (5) in Chapter 2
I^*	$\kappa_I I^* = \beta^* \tilde{J}^{V,*}$	(19)
$\tilde{C}^*$	$\tilde{Y}(s^*) = \tilde{C}^* + \frac{\kappa_I}{2} (I^*)^2 + \tilde{\gamma}_V V^*$	Equation (8) in Chapter 2
ι^*	$\tilde{W}^{S,*} = \tilde{W}^{R,*}$	Interior entry condition

If agglomeration productivity is high enough that search dominates rural residence at low entry, but search costs dominate when θ^* becomes arbitrarily small, the intermediate value theorem gives a unique interior $\iota^* \in (0, 1 - s^*)$ satisfying $\tilde{W}^{S,*} = \tilde{W}^{R,*}$.

Remark 2 (Equilibrium multiplicity and selection). Multiple equilibria are a standard feature of search-and-matching models with strategic complementarities (Diamond, 1982; Moen et al., 2021): when expected urban productivity and growth are low, agents do not enter the search pool and developers hold back, making the pessimistic outcome self-fulfilling. At the baseline calibration, the free-entry residual $\tilde{W}^S(s) - \tilde{W}^R(s)$ has two roots: a low-urbanisation trap near $s \approx 0.1$ and the calibrated equilibrium at $s^* = 0.85$. The high- s root is selected because it is the only root consistent with the UK calibration target, and the residual is increasing through zero at the low- s root, making it a repeller.

A.3 Planner Conditions and Welfare Expressions

For the baseline exponential matching function, writing $x \equiv \mu\theta^{1-\eta}$ gives

$$\varepsilon(\theta) = \frac{(1 - \eta)x}{e^x - 1},$$

which is strictly decreasing from $1 - \eta$ toward zero. The derivation of the Hosios condition stated in Chapter 3 follows. On the interior BGP, $\tilde{W}^{S,*} = \tilde{W}^{R,*}$. Subtracting (14) from (13) implies

$$\tilde{\chi} = \beta^* \pi^* (\tilde{W}^{U,*} - \tilde{W}^{S,*}) = \beta^* m(\theta^*) (\tilde{W}^{U,*} - \tilde{W}^{S,*}),$$

where the second equality uses the unconstrained branch $\pi^* = m(\theta^*)$. Nash sharing implies

$$\begin{aligned} \tilde{W}^{U,*} - \tilde{W}^{S,*} &= \phi \mathcal{S}^*, \\ \mathcal{S}^* &\equiv (\tilde{W}^{U,*} - \tilde{W}^{S,*}) + (\tilde{J}^{F,*} - \tilde{J}^{V,*}), \end{aligned}$$

so the decentralised marginal-entry condition is

$$\tilde{\chi} = \beta^* m(\theta^*) \phi \mathcal{S}^*.$$

A planner holding vacancies fixed recognises that an additional searcher changes tightness because $\theta = V/u$. Using $M = um(\theta)$,

$$\left. \frac{\partial M}{\partial u} \right|_V = m(\theta) + um'(\theta) \frac{\partial \theta}{\partial u} = m(\theta) - \theta m'(\theta) = m(\theta)[1 - \varepsilon(\theta)].$$

The planner's marginal-entry condition is therefore

$$\tilde{\chi} = \beta^* m(\theta^*) [1 - \varepsilon(\theta^*)] \mathcal{S}^*.$$

Comparing the decentralised and planner conditions yields the condition.

The numerical planner benchmark chooses an interior BGP allocation directly rather than solving a full transition problem. It solves

$$\max_{s \in (0,1), \theta > 0} \mathcal{V}(s, \theta) \equiv \frac{\mathcal{W}(s, \theta)}{1 - \beta(1 + \nu s^\xi)}, \quad (20)$$

subject to the stationary identities

$$\begin{aligned} M &= \lambda s, & u &= \frac{M}{m(\theta)}, & V &= \theta u, \\ H &= s + V, & I &= \delta_h H, & \iota &= u - \lambda s, \end{aligned}$$

the feasibility restrictions

$$\beta(1 + \nu s^\xi) < 1, \quad 0 \leq \iota \leq 1 - s, \quad q \equiv \frac{M}{V} < 1,$$

and flow welfare

$$\mathcal{W}(s, \theta) \equiv \tilde{Y}(s) - \frac{\kappa_I}{2} I^2 - \tilde{\gamma}_V V - \tilde{\chi} u - \tilde{\phi}_0 s^2.$$

Let $\theta = \theta(s)$ denote the planner's inner profile and $\mathcal{W}^*(s) \equiv \mathcal{W}(s, \theta(s))$. The outer optimum satisfies

$$\frac{d\mathcal{W}^*(s_{SP}^*)}{ds} = -\Omega^g(s_{SP}^*), \quad (21)$$

using the growth wedge defined in Chapter 3.

For the numerical comparison below, write $\hat{\theta}(s)$ for the planner's inner solution and reserve $\theta^*(s)$ for the decentralised BGP reconstruction.

Assumption 1 (Local tightness equivalence). For each s in a neighbourhood of s_{CE}^* , the decentralised tightness profile $\theta^*(s)$ coincides with the planner's inner solution $\theta(s) = \arg \max_{\theta} \mathcal{W}(s, \theta)$.

Remark 3. In a standard DMP model, the Hosios condition aligns private and efficient tightness by driving the value of vacancy creation to zero. Here housing is durable and the value of a vacant unit capitalises future returns, so level equivalence is too strong as a numerical description of the baseline. Table 7 documents the level discrepancies, the reconstruction fragility near the matching-cap boundary, and the derivative-level robustness check. Proposition 2 therefore relies on the latter sign-preserving robustness rather than literal equality of the two tightness profiles.

Table 7. Tightness profile comparison: decentralised reconstruction versus planner inner solution

s	$\theta^*(s)$	$\hat{\theta}(s)$	$ \Delta\theta $	CE feasible
0.750	0.714951	0.485692	0.229259	Yes
0.775	0.614674	0.477610	0.137064	Yes
0.800	0.520228	0.469816	0.050412	Yes
0.825	0.431891	0.462293	0.030402	Yes
0.850	0.350000	0.455027	0.105027	Yes
0.875	–	0.448005	–	No
0.900	–	0.545523	–	No
0.925	–	0.886371	–	No
0.950	–	1.640580	–	No

Note. The maximum absolute discrepancy over feasible rows is 0.229 and the mean is 0.110. The default CE reconstruction becomes numerically fragile above $s \approx 0.855$; finer bracketing places the feasibility limit around $s \approx 0.864$ at baseline parameters. At the derivative level, substituting the planner inner profile gives $D(s_{CE}^*) = 0.3666$ instead of 0.3631, preserving the positive sign with a shift of 0.0035.

Proposition 2 (Under-urbanisation). Suppose the Hosios condition holds at the decentralised equilibrium, Assumption 1 is valid locally, or the weaker sign-preserving derivative-level robustness in Table 7 holds near s_{CE}^* , $\nu > 0$, and the planner's objective is strictly concave near the decentralised BGP with an interior maximiser s_{SP}^* . Then

$$s_{CE}^* < s_{SP}^*, \quad g_{CE}^* = \nu(s_{CE}^*)^\xi < \nu(s_{SP}^*)^\xi = g_{SP}^*.$$

The associated welfare gap is

$$\Delta \equiv \frac{\mathcal{V}_{SP}^* - \mathcal{V}_{CE}^*}{\mathcal{V}_{CE}^*} \times 100\%.$$

Proof. Under the Hosios condition, the local search-congestion wedge is removed. Assumption 1 (or the weaker sign-preserving derivative-level robustness in Table 7) then implies that the remaining first-order difference between the planner and decentralised allocations at s_{CE}^* is the growth term $\Omega^g(s_{CE}^*) > 0$. Strict concavity of the planner's objective implies that the planner's interior maximiser lies to the right of the decentralised allocation, so the planner chooses higher urbanisation and therefore higher growth. $\square$

Proof of Proposition 1. The growth component is strictly positive,

$$\Omega^g(s_{CE}^*) = \beta \nu \xi(s_{CE}^*)^{\xi-1} \mathcal{V}^*(s_{CE}^*) > 0,$$

while at the calibration the flow-welfare slope is numerically small relative to Ω^g , as documented in Section 5.1.3 and Appendix C. The growth term therefore dominates and the calibrated wedge is positive. $\square$

A.4 Comparative Statics

Proposition 3 (Matching efficiency). *In the unconstrained regime, $\partial M_t / \partial \mu > 0$ at fixed (u_t, θ_t) , and hence $\partial s_{t+1} / \partial \mu > 0$ through the law of motion for s_t . At the calibrated BGP, general-equilibrium feedbacks preserve the sign numerically, so*

$$\frac{\partial g^*}{\partial \mu} = \nu \xi(s^*)^{\xi-1} \frac{\partial s^*}{\partial \mu} > 0.$$

Proof. With the baseline matching specification $m(\theta) = 1 - \exp(-\mu \theta^{1-\eta})$, $\partial m(\theta) / \partial \mu = \theta^{1-\eta} \exp(-\mu \theta^{1-\eta}) > 0$ for all $\theta > 0$. Thus $M_t = u_t m(\theta_t)$ rises with μ at fixed (u_t, θ_t) , raising s_{t+1} through equation (2). The growth result follows from the chain rule applied to $g^* = \nu(s^*)^\xi$. $\square$

Proposition 4 (Construction friction). *Holding $\mathbb{E}_t[\tilde{J}_{t+1}^V]$ fixed, $\partial I_t / \partial \kappa_I = -\beta(1 + \nu s_t^\xi) \mathbb{E}_t[\tilde{J}_{t+1}^V] / \kappa_I^2 < 0$. At the calibrated BGP, the induced general-equilibrium response satisfies $\partial H^* / \partial \kappa_I < 0$ and $\partial s^* / \partial \kappa_I < 0$ numerically.*

Proof. Differentiate equation (6) in Chapter 2 with respect to κ_I , holding the continuation value fixed. Since $H_{t+1} = (1 - \delta_h)H_t + I_t$, lower investment depresses future housing capacity and, in the calibrated equilibrium, lowers the stationary housed share. $\square$

Proposition 5 (Vacancy carrying cost). *Let $\rho \equiv \beta^*(1 - \delta_h)$. On the BGP,*

$$\tilde{J}_{V,*}^{V,*} = \frac{-\tilde{\gamma}_V + \rho q^* \tilde{J}_{F,*}^{F,*}}{1 - \rho(1 - q^*)}, \quad \frac{\partial \tilde{J}_{V,*}^{V,*}}{\partial \tilde{\gamma}_V} = \frac{-1}{1 - \rho(1 - q^*)} < 0.$$

where the derivative holds at fixed $(q^, \tilde{J}_{F,*}^{F,*})$. Consequently, higher vacancy carrying costs reduce investment locally through (19), and at the calibrated BGP also reduce H^* and s^* numerically.*

Appendix B

Calibration Details and Parameter Sensitivity

This appendix documents the closed-form calibration step used in Chapter 4. All numerical tables below are generated directly from the MATLAB code.

B.1 Closed-Form Recovery of Internal Parameters

The calibration fixes the following three target moments:

$$s^* = 0.85, \quad \frac{V^*}{H^*} = 0.07, \quad \theta^* = 0.35.$$

The remaining primitives are treated as externally set. Given the target vacancy share,

$$H^* = \frac{s^*}{1 - V^*/H^*}, \quad V^* = H^* - s^*, \quad u^* = \frac{V^*}{\theta^*}.$$

With $\lambda \equiv \delta_h + (1 - \delta_h)\delta$, stationarity of the housed stock implies

$$M^* = \lambda s^*, \quad q^* = \frac{M^*}{V^*}, \quad \pi^* = \frac{M^*}{u^*}, \quad I^* = \delta_h H^*.$$

The growth-adjusted discount factor and housing-survival discount term are

$$\beta^* \equiv \beta(1 + \nu(s^*)^\xi), \quad \rho \equiv \beta^*(1 - \delta_h).$$

The second definition matches the notation used in Appendix A.

B.1.1 Matching efficiency

Under the baseline matching function $m(\theta) = 1 - \exp(-\mu\theta^{1-\eta})$, the tightness target identifies

$$\mu = -\frac{\log(1 - \pi^*)}{(\theta^*)^{1-\eta}}.$$

B.1.2 Agglomeration productivity

Interior free entry implies $\tilde{W}^{S,*} = \tilde{W}^{R,*}$ and hence

$$\tilde{W}^{R,*} = \frac{1}{1 - \beta^*}.$$

Using the search Bellman equation,

$$\tilde{W}^{U,*} - \tilde{W}^{R,*} = \frac{\tilde{W}^{R,*} - (1 - \tilde{\chi}) - \beta^*(1 - \pi^*)\tilde{W}^{R,*}}{\beta^*\pi^*} - \tilde{W}^{R,*}.$$

Denote this urban surplus gap by $\Delta W^* \equiv \tilde{W}^{U,*} - \tilde{W}^{R,*}$. Then the housed Bellman equation implies

$$z_0 = \frac{\tilde{W}^{U,*} - \beta^* [(1 - \lambda)\tilde{W}^{U,*} + \lambda\tilde{W}^{R,*}] + \tilde{r}^* + \tilde{\phi}_0 s^*}{(s^*)^\psi}.$$

B.1.3 Construction cost

Given q^* and ΔW^* , the landlord block is linear in the stationary rent:

$$\begin{aligned}\tilde{J}^{F,*} &= \tilde{r}^* + \rho[(1 - \delta)\tilde{J}^{F,*} + \delta\tilde{J}^{V,*}], \\ \tilde{J}^{V,*} &= -\tilde{\gamma}_V + \rho[q^*\tilde{J}^{F,*} + (1 - q^*)\tilde{J}^{V,*}], \\ \phi(\tilde{J}^{F,*} - \tilde{J}^{V,*}) &= (1 - \phi)\Delta W^*.\end{aligned}$$

Solving this system yields $\tilde{r}^*$ and $\tilde{J}^{V,*}$, and the investment first-order condition then identifies

$$\bar{\kappa}_I = \frac{\beta^* \tilde{J}^{V,*}}{I^*}.$$

This is the exact inversion implemented in `calibrate_bgp_targets.m`; no nonlinear solver is used at the calibration stage. The replication package's `verify_bgp_consistency.m` plugs the calibrated values back into all structural model equations and confirms that every residual is below 10^{-10} .

B.2 Implied Baseline BGP Quantities

Table 8 reports the BGP quantities implied by the baseline calibration.

Table 8. Implied BGP quantities at the baseline calibration

Object	Value	Object	Value	Object	Value
λ	0.0690	H^*	0.9140	V^*	0.0640
u^*	0.1828	M^*	0.0587	π^*	0.3208
q^*	0.9167	ι^*	0.1241	I^*	0.0183
g^* (% p.a.)	1.2400	μ	0.6540	z_0	1.1525
$\bar{\kappa}_I$	51.4529	$\varepsilon(\theta^*)$	0.4095	Hosios gap	-0.0905

B.3 Calibration Sensitivity to the Growth Exponent

Chapter 4 treats ξ as externally set. Holding the calibration targets fixed and re-running the closed-form inversion shows that changing ξ mainly affects the growth rate and the recovered construction-cost parameter, while μ and z_0 barely move at the reported precision. Table 9 reports the full sweep.

Table 9. Calibration recovery as ξ varies

ξ	g^* (% p.a.)	$\bar{\mu}$	z_0	$\bar{\kappa}_I$	β^*
0.50	1.2810	0.6540	1.1525	51.8945	0.9723
0.60	1.2603	0.6540	1.1525	51.6710	0.9721
0.70	1.2400	0.6540	1.1525	51.4529	0.9719
0.80	1.2200	0.6540	1.1526	51.2400	0.9717
0.90	1.2003	0.6540	1.1526	51.0322	0.9715
1.00	1.1810	0.6540	1.1526	50.8293	0.9713

B.4 Calibration Sensitivity to Auxiliary Parameters

The three auxiliary parameters $\tilde{\gamma}_V$, $\tilde{\chi}$, and $\tilde{\phi}_0$ are not pinned by the target moments. Table 10 reports one-at-a-time sweeps around the baseline values, again holding the three target moments fixed and recalibrating the internally recovered parameters each time.

Table 10. One-at-a-time auxiliary-parameter sweeps

Parameter	Value	μ	z_0	$\bar{\kappa}_I$
$\tilde{\gamma}_V$	0.0050	0.6540	1.1576	57.0456
$\tilde{\gamma}_V$	0.0100	0.6540	1.1525	51.4529
$\tilde{\gamma}_V$	0.0150	0.6540	1.1475	45.8602
$\tilde{\gamma}_V$	0.0200	0.6540	1.1425	40.2675
$\tilde{\chi}$	0.0100	0.6540	1.1182	20.1337
$\tilde{\chi}$	0.0200	0.6540	1.1525	51.4529
$\tilde{\chi}$	0.0300	0.6540	1.1869	82.7721
$\tilde{\chi}$	0.0400	0.6540	1.2213	114.0912
$\tilde{\phi}_0$	0.0500	0.6540	1.1097	51.4529
$\tilde{\phi}_0$	0.1000	0.6540	1.1525	51.4529
$\tilde{\phi}_0$	0.1500	0.6540	1.1954	51.4529
$\tilde{\phi}_0$	0.2000	0.6540	1.2382	51.4529

Note. The omitted columns from the generated sweep table are invariant at the reported precision: $g^* = 1.2400\%$, $\pi^* = 0.3208$, $q^* = 0.9167$, and $\iota^* = 0.1241$.

Appendix C

Welfare Diagnostic Robustness

This appendix reports three robustness exercises for the local welfare diagnostic $D(s_{CE}^*)$ defined in Chapter 3. Each exercise holds the calibration targets $(s^*, V^*/H^*, \theta^*)$ fixed and recalibrates the internal parameters $(\mu, z_0, \bar{\kappa}_I)$ via the closed-form inversion of Appendix B.

C.1 Bargaining Weight

Table 11 varies the tenant bargaining weight ϕ over $[0.30, 0.70]$, spanning both sides of the Hosios-efficient value $1 - \varepsilon(\theta^*) \approx 0.591$. The diagnostic remains strictly positive throughout, ranging from 0.327 to 0.447, so the headline welfare conclusion does not depend on the symmetric-Nash baseline.

Table 11. Sensitivity of the local welfare diagnostic to ϕ

ϕ	Hosios gap	$d\widetilde{W}^*/ds$	Ω^g	$D(s_{CE}^*)$	Interior BGP
0.30	-0.2905	0.0659	0.3814	0.4473	✓
0.35	-0.2405	0.0429	0.3743	0.4172	✓
0.40	-0.1905	0.0256	0.3691	0.3947	✓
0.45	-0.1405	0.0122	0.3650	0.3771	✓
0.50	-0.0905	0.0014	0.3617	0.3631	✓
0.55	-0.0405	-0.0074	0.3590	0.3516	✓
0.60	0.0095	-0.0147	0.3568	0.3421	✓
0.65	0.0595	-0.0208	0.3549	0.3341	✓
0.70	0.1095	-0.0261	0.3532	0.3272	✓

Note. Hosios gap is $\phi - [1 - \varepsilon(\theta^*)]$. Calibration targets $(s^*, V^*/H^*, \theta^*) = (0.85, 0.07, 0.35)$ are held fixed across rows; internal parameters $(\mu, z_0, \bar{\kappa}_I)$ are recalibrated at each value of ϕ .

C.2 Agglomeration Curvature

Table 12 varies the agglomeration curvature ψ over $[0.02, 0.08]$, the range spanned by the empirical agglomeration literature (Rosenthal and Strange, 2004; Combes and Gobillon, 2015). The diagnostic remains strictly positive, ranging from 0.327 to 0.399. The near-invariance is expected: once $(s^*, V^*/H^*, \theta^*)$ are fixed, ψ affects $D(s_{CE}^*)$ only through the calibrated agglomeration kernel rather than through the growth wedge directly.

Table 12. Sensitivity of the local welfare diagnostic to ψ

ψ	z_0	Hosios gap	$d\tilde{W}^*/ds$	Ω^g	$D(s_{CE}^*)$	Interior BGP
0.02	1.1469	-0.0905	-0.0346	0.3617	0.3271	✓
0.03	1.1488	-0.0905	-0.0226	0.3617	0.3391	✓
0.04	1.1507	-0.0905	-0.0106	0.3617	0.3511	✓
0.05	1.1525	-0.0905	0.0014	0.3617	0.3631	✓
0.06	1.1544	-0.0905	0.0134	0.3617	0.3751	✓
0.07	1.1563	-0.0905	0.0254	0.3617	0.3871	✓
0.08	1.1582	-0.0905	0.0374	0.3617	0.3991	✓

Note. Calibration targets are held fixed across rows; z_0 and the other internal parameters are recalibrated at each value of ψ . The baseline row $\psi = 0.05$ reproduces Table 5.2.

C.3 Matching-Function Specification

Table 13 compares the baseline exponential matching function against a Cobb-Douglas specification, recalibrating μ in each case to hit $\theta^* = 0.35$. The diagnostic is 0.363 under the exponential baseline and 0.357 under Cobb-Douglas, a difference below 2%.

Table 13. Matching-function robustness

Specification	π^*	q^*	μ	$\varepsilon(\theta^*)$	Hosios gap	$d\tilde{W}^*/ds$	Ω^g	$D(s_{CE}^*)$
Exponential (baseline)	0.3208	0.9167	0.6540	0.4095	-0.0905	0.0014	0.3617	0.3631
Cobb-Douglas	0.3208	0.9167	0.5423	0.5000	0.0000	-0.0045	0.3617	0.3572
Log kernel	0.3208	0.9167	0.6540	0.4095	-0.0905	0.0019	0.3617	0.3636

Note. All rows are calibrated to the same tightness target $\theta^* = 0.35$. The baseline exponential function bounds match probabilities in $[0, 1]$ globally; Cobb-Douglas recovers the same stationary allocation over the admissible interior branch. The log-kernel row retains the exponential matching function and replaces the agglomeration term with $z_0(1 + \psi \log s)$.

Summary

Across all three exercises, the local welfare diagnostic remains strictly positive. The bargaining-weight sweep is the substantive robustness test; the agglomeration curvature and matching-function checks rule out parameter-envelope and functional-form artefacts. The agglomeration kernel specification is similarly inconsequential: replacing the power-law form $z_0 s^\psi$ with a log specification shifts $D(s_{CE}^*)$ by 0.0005, confirming the diagnostic is not an artefact of the kernel's functional form. The flow slope $d\tilde{W}^*/ds$ is evaluated by a centred finite difference at $h = 10^{-5}$; variation across $h \in \{10^{-3}, \dots, 10^{-7}\}$ is below 10^{-4} , confirming the diagnostic is not sensitive to the discretisation choice.

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Use of AI Tools

AI assistance was used in two limited capacities during this project. In all cases, the model's central idea, economic mechanisms, mathematical derivations, calibration choices, and final wording are my own.

1. **Computational support.** Claude Opus 4.6 was used to debug and organise MATLAB code implementing the BGP solver, welfare diagnostic, and impulse response routines. Typical prompts concerned translating structural model conditions into code checks and diagnosing errors in welfare-block outputs. All suggestions were reviewed and adapted before incorporation; no code was accepted without verification against the model's analytical conditions.
2. **Adversarial theory checking.** At selected points I used Claude Opus 4.6 as a critical interlocutor to probe whether stated propositions admitted counterexamples or whether knife-edge assumptions were doing unacknowledged work. This use was deliberately adversarial. The prompts asked the model to find problems rather than generate content. Any resulting clarifications were verified against formal derivations before inclusion. The model's failure to construct a counterexample was treated as weak evidence only; the formal proofs in Appendix A are my own.

Records of relevant exchanges are retained for verification if required. This disclosure is made in compliance with the Department's stated rules on permitted AI use.